

Human-Computer Interaction

Experimental Design

+ Measurement Basics

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Today's Agenda

- Part 1: *Experimental research basics*
- Part 2: *A step-by-step design guide*
- Part 3: *Measurement basics*
- Hands-on: *reverse-engineer an experiment*

Due today, 9:30 am

- **Qualitative Field-Method Plan** (*individual*)
- Your team's **3-line project status**: done / blocker / next

Reading for today

Lazar, Feng & Hochheiser (2017), *Research Methods in Human-Computer Interaction*

- **Ch. 2:** Experimental research
- **Ch. 3:** Experimental design

Looking ahead to the Oct 16 crit: Ch. 5 (Surveys) and Ch. 13 (Measuring the human).

Part 1: Experimental Research Basics

What is a hypothesis?

A testable statement of the predicted relationship between at least two variables, answering a *research question*.

Question: How does seeing an AI's answer before trying a problem affect what students learn?

Hypothesis: Students who try a problem before seeing the AI's answer will score higher on a later test, taken without AI, than students who see the AI's answer first.

Two features: **falsifiable** and **testable**.¹

¹Popper, K. (1959). *The Logic of Scientific Discovery*. Hutchinson.

Good hypotheses are

1. **Testable**: you can manipulate IVs and measure DVs
2. **Falsifiable**: the data could prove them wrong
3. **Parsimonious**: the simplest form of the prediction
4. **Precise**: operationalizable
5. **Useful**: informs existing or new theory

Five types of variable²

Type	What it is	In our AI-first example
Independent (IV)	what is manipulated	try first vs. see the AI's answer first
Dependent (DV)	what is measured	later test score, without AI
Control	what is held constant	same problems, same time limit
Random	what is allowed to vary randomly	which students, which problems
Confounding	what correlates with the IV or DV	prior knowledge, if groups differ

² McGrath, J. E. (1995). Methodology matters. In *Readings in HCI* (pp. 152-169). Morgan Kaufmann.

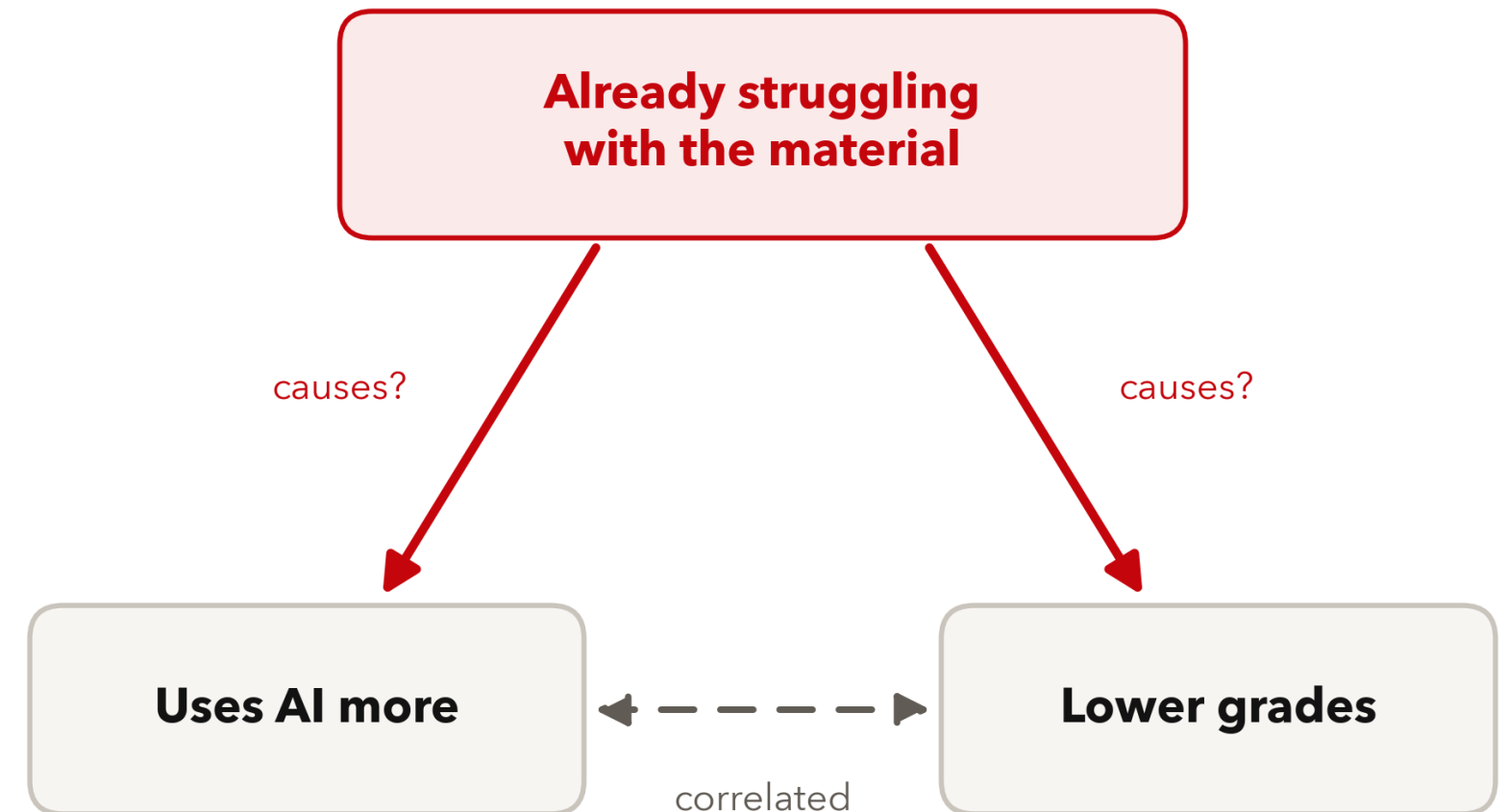
What relationships are we after?

- **Causal:** one variable (DV) depends on and is affected by another (IV)
- **Correlational:** two variables move together, possibly driven by a third

Correlation $\neq$ causation.

A correlation, and a third variable

Hypothetical: students who use AI more get lower grades



The AI may cause nothing: a common cause can drive both.

Three questions → three designs

Know all the variables?	Control the IV?	Measure them?	Design	Example
Yes	Yes	Yes	True experiment	randomly assign try-first vs. AI-first
Yes	No	Yes	Quasi-experiment	compare daily AI users with rare users
No	Yes or no	Yes	Correlational	survey AI use and grades

Factors, levels, assignment

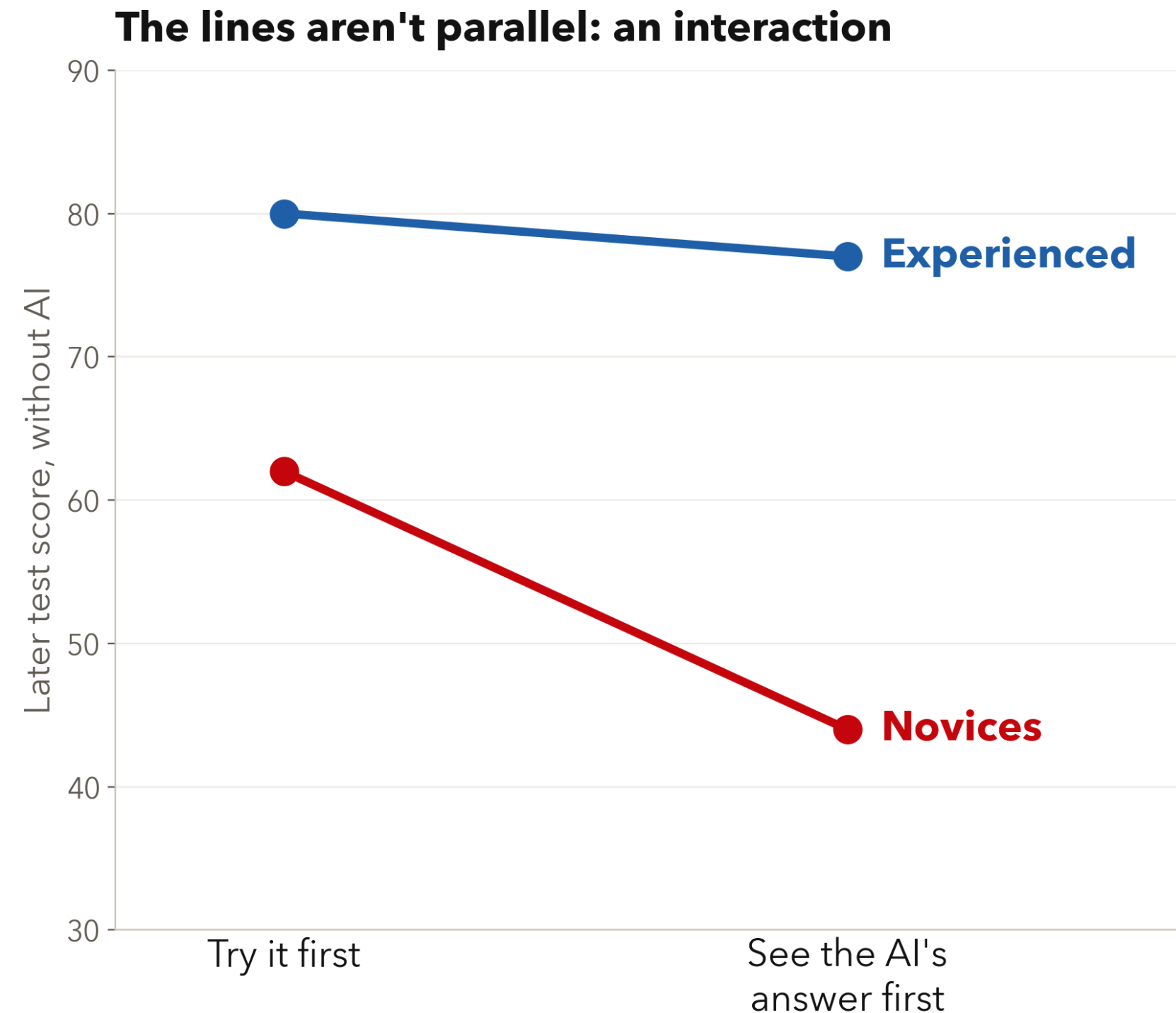
IVs are **factors**; their values are **levels**. E.g., factor *AI timing*, levels *try first* and *AI first*.

	Within-participants	Between-participants
Who sees what	Everyone sees every level	Each group sees one level
Strength	More statistical power, fewer people	Nothing carries over between levels
Weakness	Transfer: learning, fatigue	Needs more people

Multiple factors → factorial designs

Cross two or more IVs, e.g. *AI timing* (try first / AI first) × *experience* (novice / experienced).

- **Main effect:** effect of a single factor on the response
- **Interaction effect:** the effect of one factor depends on the level of another
- More efficient, but more demanding (larger n)



Illustrative data, constructed to show the effect.

Controlling for bias

Random sampling + assignment protect validity.

- **Counterbalancing**: vary the order of levels to defuse *transfer* (learning, fatigue)
 - **Blocking / Latin square**: structured counterbalancing
 - If transfer is **asymmetrical**, go *between-participants*
- **Blinding**: participants (single-blind), or participants and experimenters (double-blind), don't know the condition

Part 2: Step-by-Step Design Guide

Try it: *design your experiment, step by step*
da.gd/GH901



The seven steps

1. **Formulate** the research question
2. **Identify** variables
3. **Generate** hypotheses
4. **Determine** the experimental design
5. **Develop** task & procedure
6. **Determine** manipulations & measurements
7. **Identify** participants

Step 1: Research question

What are the effects of X on Y under conditions Z?

Variables

Target population,
experimental context, etc.

Check it against **FINER**: **F**easible · **I**nteresting · **N**ovel · **E**thical · **R**elevant⁴

⁴Farrugia et al. (2010). Research questions, hypotheses, and objectives.

Steps 2-3: Variables → hypotheses

- **Identify variables:** IVs (factors + levels), DVs, random factors
- **Hypothesize** in the canonical form:

RQ: What are the effects of X on Y under conditions Z ?

H: Y will be higher (or lower) at level 1 of X than at level 2, under Z .

- **Justify every H** from exploratory results, related theory, or reasoned face validity

Step 4: Experimental design

How participants are allocated to conditions:

- **Within-participants** (repeated measures): power, fewer n ; transfer/demand risk
- **Between-participants** (independent measures): no transfer; more n , more variance
- **Mixed-model** (split-plot): some factors within, some between

Prefer factorial (2^k) designs where feasible.

Step 5: Task & procedure

Task: a reasonable representation of the real-world context Z ; relevant, controllable, motivating.

- **Distractor tasks** add load: e.g., notifications arriving while working with the AI
- **Deceptive tasks** use a cover story: e.g., "rate the essay's clarity" when the study measures reliance on AI suggestions

Procedure: a replicable step-by-step of instructions, roles, equipment, and timeline (consent → measures → compensation).

Steps 6-7: Manipulations & participants

- **Manipulations & measurements:** IVs can be *manipulated* or *measured*; control levels give a *baseline* (no AI) or a *gold standard* (an expert human)
- **Participants:** a representative, sufficiently large, balanced sample of the target population (Z)

Everything not manipulated must be measured, which is Part 3.

Part 3: Measurement Basics

What do we measure when we measure?

Measurement: assigning a number to a characteristic so it can be compared.

- **Quantitative:** the degree of an attribute (time to finish a task, share of AI suggestions accepted)
- **Qualitative:** a description ("she rejected the suggestion because it sounded generic")

Four types of variable

From least to most precise:

- **Nominal**: categories (which assistant: none / ChatGPT / Copilot)
- **Ordinal**: ranked, unequal gaps (rank three AI drafts best to worst)
- **Interval**: equal gaps, no true zero (1–7 trust rating, treated as interval)
- **Ratio**: equal gaps, real zero (time on task, number of edits)

Four kinds of measure

1. **Objective**: against a standard (test score, task accuracy)
2. **Behavioral**: what people do (accepting or editing AI suggestions, gaze toward a robot)
3. **Subjective**: self-report (trust in the AI, perceived workload)
4. **Physiological**: from the body (eye tracking, EEG, skin conductance)

What makes a measure good?

1. **Validity**: are we measuring what we intend? (*face, construct, criterion, content, ecological*)
2. **Reliability**: is the measure consistent across repeats? ($\alpha \geq .70$ desirable)
3. **Quality**: is error controlled? (*random = noise; systematic = bias*)

Full treatment (scales, α , power) at the Oct 16 Method-Design Crit.

Hands-on: reverse-engineer an experiment (30 min)

Pods	Paper	
1-5	A · Jakesch et al. (2023)	Co-writing with opinionated LMs
6-9	B · Dhillon et al. (2024)	Scaffolding levels in co-writing
10-13	C · Buçinca et al. (2021)	To trust or to think

Abstract + intro (5 min) → method (15 min) → debrief (10 min).
Nothing to submit.

Grab the worksheet

- One copy per pod: File → Make a copy
- Nothing to submit today
- Each item maps to a step of your **Step-by-Step Experimental Design**

da.gd/ZA9eH



Debrief: three papers, three designs

	A • Jakesch	B • Dhillon	C • Buçinca
Assignment	between, random	within, Latin square	conditions randomized by block
Hypotheses	none stated	research question	explicit H1a, H1b, H2
People	1,506 online (Prolific)	131, live on Zoom	199 (Mechanical Turk)

What's Next?

- **Monday: Accessibility, Aging & AI**
 - **Read:** Wobbrock et al. (2011) · Glazko et al. (2023)
 - Post your provocation slide
- **Fri Oct 16: Method-Design Crit**
 - **Due 9:30 am:** Step-by-Step Experimental Design (*individual*)
 - Draft it with the guide (**da.gd/GH901**), then develop each part